

PRACTICAL No. 5

AIM: Study & Implementation of different types of constraints.

Theory:

CONSTRAINTS:

Constraints are used to specify rules for the data in a table. If there is any violation between the constraint and the data action, the action is aborted by the constraint. It can be specified when the table is created (using CREATE TABLE statement) or after the table is created (using ALTER TABLE statement).

SQL constraints are used to specify rules for the data in a table. They are used to limit the type of data that can be stored in a particular column of a table. Constraints ensure that the data stored is valid. This ensures the accuracy and reliability of the data in the database.

Constraints could be either at the column level or at the table level. Column level constraints are applied only to one column, whereas table level constraints are applied to the entire table.

In Oracle the integrity constraints classified into three groups:

1. Domain Integrity Constraint
2. Entity Integrity Constraint
3. Referential Integrity Constraint

1. DOMAIN INTEGRITY CONSTRAINT

a. DEFAULT Constraint

DEFAULT constraint is used to provide a default value for the fields of a table. That is, if at the time of entering new records in the table if the user does not specify any value for these fields then the default value will be assigned to them. The user needs to define these default values.

Note: Write a query to be executed in practical slot related to default constraints.

b. NOT NULL Constraint

By default, a column can hold NULL values. If we don't enter any value in a particular column, it will store NULL if the default value has not been specified. The NULL value is omitted in various operations in a database. To ensure that a column does not contain a NULL value, NOT NULL constraint is used.

NOT NULL constraint applied to a column ensures that the particular column always contains a value, i.e., we cannot insert a new record, or update a record without providing a value for that field.

A NULL is not the same as no data, blank, or a zero-length character string such as ' ', rather, it represents unknown data or that no data entry has been made.

Note: Write a query to be executed in practical slot related to NOT NULL constraints.

c. CHECK Constraint

CHECK constraint is used to limit the value range that can be placed in a column. Using check constraint, we can specify conditions for a field, which will be evaluated at the time of entering the data to a column. If the condition evaluates to false, the record violates the constraints and it will not be entered in the table.

If we define a CHECK constraint on a single column it allows only certain values for this column.

If we define a CHECK constraint on a table it can limit the values in certain columns based on values in other columns in the row.

Note: Write a query to be executed in practical slot related to Check constraints.

2. ENTITY INTEGRITY CONSTRAINT

a. UNIQUE Constraint

UNIQUE constraint ensures that all the values stored in a column are different from each other. The UNIQUE Constraint prevents two records from having identical values in a column, i.e., the values must not be repeated.

We can use UNIQUE constraint in a single column or multiple columns.

UNIQUE constraint allows columns to have NULL values.

Note: Write a query to be executed in practical slot related to unique constraints.

b. PRIMARY KEY Constraint

A primary key constraint uniquely identifies each row/record in a database table. Primary keys must contain unique values. A primary key column cannot have NULL values.

A table can have only one primary key, which may consist of single or multiple fields. When multiple fields are used as a primary key, they are called a composite key.

The field with a PRIMARY KEY constraint can be used to uniquely identify each record of a table in various database operations.

Note: Write a query to be executed in practical slot related to primary constraints.

3. REFERENTIAL INTEGRITY CONSTRAINT

FOREIGN KEY Constraint

Foreign Key Constraint is used to link two tables. It is used to establish a relationship between the data in two tables.

A Foreign Key is a column or a combination of columns whose values match a Primary Key in a different table. The table containing the FOREIGN KEY is called the child table, and the table containing the candidate key is called the parent table.

FOREIGN KEY is used to enforce referential integrity. It is used to prevent actions that would destroy links between tables. It also prevents invalid data from being inserted into the foreign key column, because it has to be one of the values contained in the table it points to.

Note: Write a query to be executed in practical slot related to foreign key constraints.

Result:

Thus, we study and implement the different types of constraints in SQL.